

Honours Report

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This report is divided into two parts:

1. **Quantitative work:** This contains information about all the NLP work undertaken during the semester and all the neural NER models that were trained.
2. **Qualitative work:** This part will talk about the various literature that I have read as part of my research and will give an overview of the ideas presented in each of the books.

Quantitative Work

Problem Statement

This research project aims to delve into the intricacies of historical exchanges involving commodities and human movement within the Western Himalayas. The overarching objective is to comprehend the patterns that have shaped the region's socio-economic landscape throughout history. A corpus consisting of approximately 20 meticulously selected documents has been assembled to undertake this investigation, offering valuable insights into the past. The pivotal task at hand involves training a model capable of effectively extracting essential entities, such as place names, personal names, and commodities, from these textual sources. Furthermore, these entities will be geospatially mapped onto cartographic representations to elucidate their geographical distribution and connections visually. After this, the task was to figure out the context in which these entities were being mentioned in the texts and to try and figure out their patterns of movement. Overall, we wish to come up with an end-to-end pipeline that can generate cartographic maps from historical texts, which will provide historians with rich insights into the region and aid them in their qualitative analysis.

Dataset Used

To train the NER model, a dataset was carefully crafted over the summer, which categorised tokens into coarse-grained and fine-grained entities. For now, we only used the coarse-grained entities (Place names, people's names, and commodities) to train our NER model, as the number of data points for every fine-grained entity was too small to deliver workable results. In total, we have about 6000 data points for the three coarse-grained entities.

Here is a link to our dataset: [Mapping Himalayas Historical Texts, Annotated Data.xlsx](#)

This is what the dataset looks like:

Geographical Entities (A)							People (B)					Commodities (C)									
Habitations	Rivers, Lakes	Peaks	Passes	Forests	Paths, Roads	Other Location of Note	Private Individuals	Groups	Official Positions	Socio-Cultural Status Position	Religious Persons	Salt	Borax	Sheep Wool	Yak Wool	Pushm White	Pushm Black	Cannabis	Food	Cloth	
	1	2	3	4	5	6	7	1	2	3	4	5	1	2	3	4	5	6	7	8	9
Haridwar	Jumna, Jamna, Yamuna, Yumna					Cangra	Lieutenant Webb	Cafilahs	Lieutenant	Surya-Vanshi	Ram-Rae	Rock salt		Shawls	Shawls				asafoetida	Dutas	
Badrinath	Ganga, Ganges, Emodious					Sansar Dhara	Puki Sonar	Hindus	Raja	Gosains, Sannyasis	Daulet Rao Sinda's			Loi					Apricots	Pattus	
Srinagar	Yamuna, Jumna					Mahadevaca-linga	Harbalam	Delals	Jemadar	Har-Sewac-Ram	Raja Iswara								Figs	Chiras	
Kedarnath	Mansarobar					Baidya Nath	Goman-Sing-Cum-ar	Paharis		Rajputs	Raja Carna								Prunes	Baflas	
Devaprayag	Alcananda, Dawli, Vishnu Ganga					Rameswara	Bhaira Tapah	Casias		Udasis	Dhocal Gir								Raisins	Muslin	
Almora	Cedar					Indra Dhara	Bhim Sa'a	Nat's		Jogis, Camp'hatai	Dayaram Patha								Almonds	Mashru, Sarcenet	
Bareilly	Satlej					Masret Chawra	Har-Sewac-Ram	Tartars		Zemindars									Pistachio Nuts	Kharua	
Canachala, Kankhal	Bhagirathi					Berham Deo-ca-Nau	Shista Tapah	Butias		Fakirs									Pomegranates		
Seharanpur	Nagar					Cala Mandi	Buddi Tapah	Beoparis		Brahmins									Cocoa-nuts		
Carcari	Retal Nadi					Lalmandi	Parsuram Tapah	Musleman		Rajputs											
Hrishicesa, Hrik'hikes	Suar Nadi					Bhimeswara	Amer Sing Cadzi, Amer Singh	Namburi tribes		Tapahs											
Nagal	Cajani Nadi					Chicata Ghari	Raja Sansar Chand	Sadhucaris		Chawtras											
Mugra	Billang					Har-ca-Pairi	Rudravir Sa'a	Pandus		Rauhil											
Bohan Devi	Beyah, Hyphasis					Gangautri, Gangotri	Ghur Ban Judh Bicrama Sa'hi	Tappas		Bairagis											
Lullari	Cali Ganga, Mandacini						Bhim Sin'h Tapah	Mewatts		Sahucars											
Haddari	Pindar						Upendra Sahi			Baniah											

Methodology

Now, we come to the central part of training the NER model. For this, I tried multiple approaches throughout the first two months of my semester.

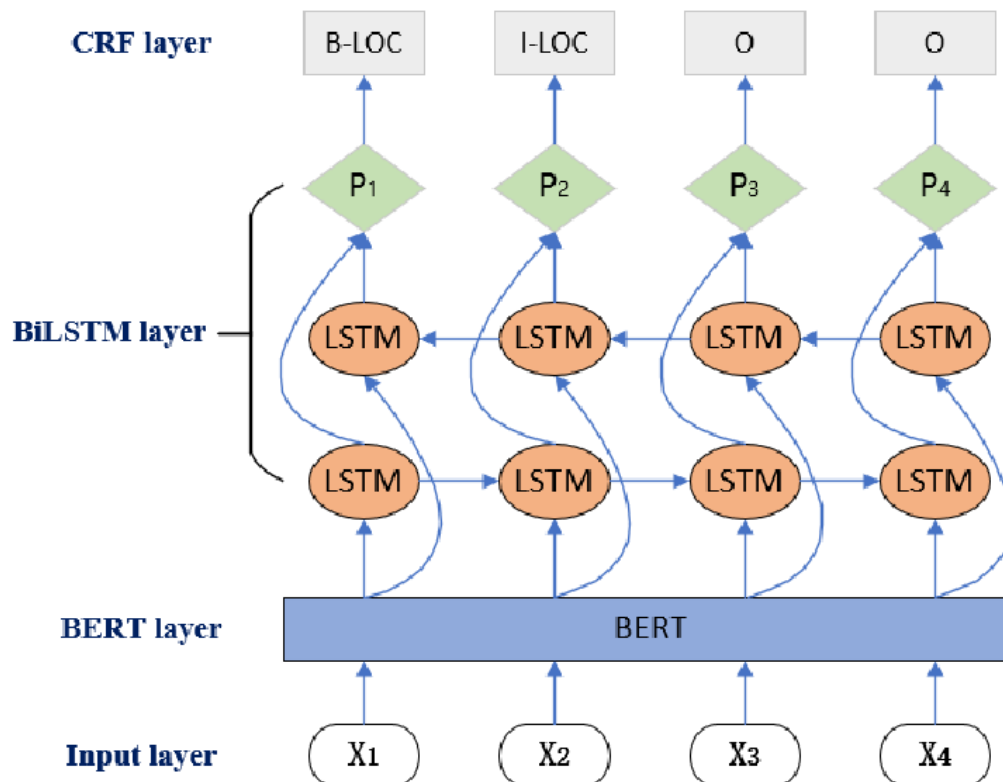
The first main idea was to make use of **transfer learning** to train the model. Transfer learning is a machine learning technique that leverages knowledge gained from solving one problem and applies it to a different but related problem. In the context of Natural Language Processing (NLP), transfer learning has proven to be particularly effective in training Named Entity Recognition (NER) models on historical texts. Historical texts often pose unique challenges for NER tasks, such as archaic language, varied writing styles, and obsolete terminology. Traditional NER models may struggle to perform well on these datasets due to the lack of specific training examples and the inherent differences in language use over time. Transfer learning addresses this issue by pre-training a model on a large and diverse dataset, often composed of modern language. During this pre-training phase, the model learns general language patterns, contextual information, and semantic relationships. Once pre-trained, the model can be fine-tuned on a smaller dataset of historical texts to adapt its knowledge to the specific characteristics of the target domain.

Our idea was to employ foundational Large Language Models (LLMs) like BERT and RoBERTa to train the NER model. Since these models have been trained on a large amount of dataset, we want to use (or transfer) their knowledge of the language to our classification model. Thus, the architecture we trained was a **BERT-BiLSTM-CRF** and a **RoBERTa-BiLSTM-CRF** architecture. BiLSTM is a type of recurrent neural network (RNN) that processes input sequences bidirectionally. It consists of two LSTM layers—one

processing the input sequence from left to right and the other from right to left. This bidirectional processing helps capture long-range dependencies and contextual information from both directions, complementing the capabilities of BERT. CRF is a probabilistic graphical model that considers the dependencies between neighbouring labels in a sequence. In the context of NER, CRF is used to model the transition probabilities between different entity labels, ensuring that the predicted labels form coherent sequences. This helps improve the overall consistency and accuracy of the predicted named entities. According to previous literature on the topic, this architecture usually gave the best results.

The parameters of the LLMs (BERT/RoBERTa) are kept frozen, and only the Bi-LSTM and the CRF (the classification layer) are trained. This is because our dataset is small, and if we try to change the weights of the foundational models, we will completely mess them up, and we won't be able to learn anything. Thus, we simply borrow these models' weights (or the learnings) and use them to train our NER classification model. Freezing the BERT parameters also reduces the size of our model, which means that we can fit it easily on our Ada provided to us by the college, and the training is also quicker. **Eight different models** with the same architecture but with different hyperparameters were trained. Each of them took **about 30 hours** to train. Hence, as we can see, the process of model training was an arduous task. A separate but related approach that I tried also included the **gradual unfreezing of layers**. Instead of freezing all the layers of the large language model, we gradually unfreeze the top-most task-specific layers (Anna Rogers) and train them on a very low learning rate. This is said to guide our LLM towards our specific domain and give better results for the dataset that we are using.

This is what a BERT-BiLSTM-CRF architecture looks like:



Here are the parameters of the various models trained:

S.No.	Layers	Learning Rate	d1 (Bi-LSTM)	d2 (Bi-LSTM)	Batch Size	Epochs
1	Frozen	1e-3	128	32	4	200
2	Frozen	1e-3	128	16	8	200
3** (best)	Frozen	1e-4	128	32	8	200
4	Frozen	1e-4	256	64	4	200
5	Frozen	5e-3	256	64	4	200
6	Frozen	5e-3	128	32	8	200
7	Gradually unfrozen	1e-4	256	64	4	200
8	Gradually unfrozen	1e-4	128	32	8	200

However, this approach did not give us desirable results. Here is a **sample classification report for one of the models:**

```

classification report:
      precision    recall  f1-score   support

   O       0.95      0.99      0.97      73063
  GEO       0.45      0.12      0.19      2846
  PEO       0.53      0.10      0.16       997
  CMD       0.70      0.50      0.58       196

 accuracy              0.95      77102
 macro avg       0.66      0.43      0.48      77102
 weighted avg     0.93      0.95      0.93      77102

```

As we can see, we get very low F1 and recall scores for our 3 entities, which implies that this model cannot be used for real-world data.

The second approach that I tried was **Domain Adaptive Pretraining**.

Domain Adaptive Pretraining (DAPT) is a powerful approach in Natural Language Processing (NLP) that aims to enhance the performance of pre-trained language models, such as BERT (Bidirectional Encoder Representations from Transformers), by adapting them to specific domains. This method is beneficial when dealing with NLP tasks that involve specialised or domain-specific language, as it allows the model to learn domain-specific nuances and improve its performance on tasks within that domain.

In NLP, pretraining models on a general dataset have proven effective in capturing broad linguistic patterns. However, the language used in specialised domains can vary significantly, making it challenging for pre-trained models to perform optimally in such contexts. This is where Domain Adaptive Pretraining becomes valuable, as it enables fine-tuning on domain-specific data to better align the model with the intricacies of the target domain.

The process of Domain Adaptive Pretraining typically involves acquiring a dataset that is representative of the specific domain of interest. This dataset is then used to fine-tune the pre-trained model using tasks like Masked Language Modeling (MLM). In MLM, random words in the input text are masked, and the model is trained to predict these masked words based on the surrounding context. This process encourages the model to understand and generate language specific to the chosen domain, effectively adapting its knowledge to the intricacies of the target data.

Extending the application of Domain Adaptive Pretraining to Named Entity Recognition (NER) models for historical texts can be particularly beneficial. Historical texts often pose unique challenges, such as archaic language, obsolete terminology, and varied writing styles. By employing DAPT, a BERT-based NER model can be fine-tuned on a historical text dataset, allowing it to capture the peculiarities of language use in historical documents. This adaptation is essential for accurate entity recognition in domains where standard language models may struggle to comprehend the nuances and context-specific to historical periods.

Thus, I also employed this technique. Another domain-specific dataset of text documents was curated to be able to train our BERT model on historical text. The idea was that once the LLM has seen tokens and entities from historical documents, it will be able to classify named entities with better accuracy upon training.

Here are the parameters of the various models trained:

S.No.	Layers	Learning Rate	d1 (Bi-LSTM)	d2 (Bi-LSTM)	Batch Size	Epochs
1	Frozen	1e-3	128	32	4	200
2	Frozen	1e-3	128	16	8	200
3** (best)	Frozen	1e-4	128	32	8	200
4	Frozen	1e-4	256	64	4	200
5	Frozen	5e-3	256	64	4	200
6	Frozen	5e-3	128	32	8	200

7	Gradually unfrozen	1e-4	256	64	4	200
8	Gradually unfrozen	1e-4	128	32	8	200

The various hyperparameters are the same as in the previous case. The only change is that the domain-adapted BERT model is loaded every time we train our NER model.

The adaptive pretraining of BERT took **three days** to run on Ada. After that, I once again trained **eight different models** with separate hyperparameters for this architecture. Again, each of these models took **about 30 hours** to run. However, we still did not achieve workable results. Here is a sample **classification report**

classification report:

```

precision  recall  f1-score  support

O         0.97    0.93    0.95    73063
GEO       0.24    0.47    0.32    2846
PEO       0.21    0.18    0.19    997
CMD       0.29    0.70    0.41    196

accuracy          0.91    77102
macro avg    0.43    0.57    0.47    77102
weighted avg    0.93    0.91    0.92    77102
```

The results are **slightly better** than the ones where adaptive pretraining was not used, but they are still not good enough to be deployed and used as is.

Finally, I also tried to use the notion of **weighted loss** to improve the accuracy of my model. This is because our dataset was showcasing class imbalance, that is some entities (like 'Other') occurred much more than other entities, and thus our model was not able to learn for the other three entities well enough.

Class imbalance is a common challenge in machine learning, where certain classes have significantly fewer instances than others in a dataset. This imbalance can lead to models that are biased towards the majority class, often resulting in suboptimal performance on the minority class. Weighted loss is a method employed to address this issue by assigning different weights to classes based on their prevalence in the dataset.

Weighted loss is particularly effective in countering class imbalance because it allows the model to prioritize the learning of minority classes, ensuring that errors in these classes contribute more to the overall loss function. This adjustment helps prevent the model from being overly influenced by the majority class and encourages it to pay more attention to the minority classes during training.

The concept behind weighted loss involves assigning higher weights to the minority class and lower weights to the majority class. The weights are incorporated into the calculation of the loss function, essentially giving more importance to the underrepresented classes. This way, the model is penalized more for misclassifying instances from the minority class, making it more sensitive to the nuances of these classes.

The loss function that I used for this purpose was **Focal Loss**. Focal Loss is a specialized loss function designed to address the challenges posed by class imbalance in binary classification problems. Introduced by Lin et al. in their 2017 paper titled "Focal Loss for Dense Object Detection," the focal loss has since gained popularity, especially in tasks where one class is significantly underrepresented compared to the other. The key innovation of Focal Loss lies in its ability to down-weight the contribution of easy-to-classify examples during training, thereby focusing more on the difficult-to-classify examples. The standard cross-entropy loss treats all misclassifications equally, which can be problematic when dealing with imbalanced datasets. Focal Loss addresses this by introducing a modulating factor that reduces the loss contribution of well-classified examples, allowing the model to concentrate more on instances that are harder to classify.

Once again, the **results achieved were sub-par.**

classification report:

	precision	recall	f1-score	support
O	0.96	0.96	0.96	73063
GEO	0.27	0.28	0.28	2846
PEO	0.15	0.15	0.15	997
CMD	0.26	0.63	0.37	196
accuracy		0.92		77102
macro avg	0.41	0.50	0.44	77102
weighted avg	0.92	0.92	0.92	77102

Challenges encountered

Hence, none of the approaches we tried seemed to work. This was when we decided to seek the help of Manish sir.

One of the biggest challenges for us right now is overcoming the **OCR text**. All the documents we have are OCR documents, and when we try to read their text using python libraries, we misread many of the important words. This means that we are not able to recognise the actual entities or tokens that are present in the text. This is one of the major reasons none of our approaches for training NER was working. Thus we need to find a better OCR model that can help us read our text better with fewer errors.

Another modification that was suggested by Manish sir was to scrap the idea of using LLMs altogether and simply use a Bi-LSTM+Linear layer architecture. This is because the size of our dataset is small, and for small datasets, LLMs usually give worse performance. Thus, it is better to train a neural model with lesser number of parameters. **I have written the code for this model but have not been able to train it yet because of internal issues on Ada.** This work will be complete by end of December.

Another thing he asked us to do was to collect more text data so that we can train our NER model better. For this purpose, we shall be using the 98 volumes of the “Collected Works of Gandhi.”

Collecting more data and doing some NLP for fun

As I said, I have collected all 98 volumes of Gandhi’s collected works. For this purpose, I wrote a **scraper** that automatically downloaded all the files locally on my computer. I now have this dataset with me and will upload it to Dropbox for future reference.

Using this dataset, I also decided to build an **RAG-based chatbot** that could answer queries based on the information in these documents.

Retrieval-augmented generation refers to a paradigm in natural language processing (NLP) where a model leverages both generative and retrieval-based approaches to enhance its language understanding and content generation capabilities. This approach aims to combine the strengths of both methods, allowing the model to benefit from the richness of generative models and the precision of retrieval-based models.

In a typical retrieval-augmented generation scenario, the model has access to a pre-existing knowledge base or a database of information. When generating responses or content, the model can retrieve relevant information from this external source and incorporate it into its generated output. This helps address some of the limitations of pure generative models, such as potential inaccuracies, lack of specific details, or failure to cover a broad range of topics.

The process involves two main steps:

1. Retrieval: The model retrieves information from a predefined knowledge source. This retrieval can be based on similarity measures, semantic matching, or other techniques that identify relevant content.
2. Generation: The retrieved information is integrated into the model's generation process. This integration could involve paraphrasing, summarisation, or other techniques to ensure the retrieved content fits seamlessly with the generated output.

Thus, we shall use Gandhi's works as our database to answer related queries. **This model has been trained, and I am working on deploying it** so that everyone can use it. Such a chatbot will not only be a fun way to access Gandhi’s works but will also be an indicator into how well LLMs can perform on custom datasets and if they can generate information and answer queries to a satisfying degree.

This can also be put up on Programming Historian as a tutorial as it could be an exciting skill for many historians to learn. Learning such a methodology would allow them to analyse the vast corpus of data by simple querying and answering. While this will be slightly advanced concerning the prior knowledge required, setting up such a chatbot is not that complicated and can be conveyed easily.

Extending the previous dataset

Another task I undertook during the semester was to annotate our dataset further. The annotations were done so as to classify places as either importer, exporter, or both. Such a classification would be necessary when we eventually try to build maps and knowledge graphs, as we need to know which places are exporters of commodities and which are importers. Five documents were annotated over a week, and this dataset will be used to finetune my original NER model. The next step in this model will be to create triplets of <exporter, commodity, importer>, which will then be used for relation extraction and to correlate what commodities are being exchanged in which places.

Here is a sample of how the annotated dataset looks like:

	A	B	C
1	GEO_EXPORTER	GEO_IMPORTER	GEO_NONE
2	Chuara	Kanauwer	River Sutlej
3	Kanauwer	Chuara	Rawan Hrad
4	Kamru{salt}	Durgaon{salt}	Mmsarovar
5	Kanauwer{Neoza pine}		Ladhak
6	Kanauwer{Turnip}		Tibet
7	Leh{salt}	Kanauwer{salt}	Shipki
8	Leh{wool}	Kanauwer{wool}	Roper
9	Garu{salt}	Kanauwer{salt}	Pergunnah of Kanauwer
10	Garu{wool}	Kanauwer{wool}	Kotgerh
11	Kanauwer{grain}	Leh{grain,raisins,neozas,iron,bro	Nagkunda Pass
12	Kanauwer{grain}	Garu{grain,raisins,neozas,iron,bro	Giri River
13	Stang{salt}	Kanauwar{salt}	Jumna
14	Bekar{salt}		Jubal
15	Garu{Byangi wool}	Purbuni{Byangi wool}	Chur
16	Purbuni{barter,iron wrought, unw	Garu{barter,iron wrought, unwro	Chepal
17	Plains{Matchlock,swords}	Purbuni{Matchlocks, swords}	Dudu
18	Garu{goats,sheep}	Purbuni{goats,sheep}	Kuara
19	Majan{Goat,sheep,byangi wool}	Murang{Goat,sheep,byangi wool}	Jako
20	Shipki {Wool}	Nissang{Salt}	Durgaon
21	Shipki {Salt}	Murang {Salt}	Gunas Pass
22	Meyang {Salt}	Murang - {Wool}	Nuru Bassa
23	Meyang {Wool}	Cashmir {Shawl wool}	Nilang

Code Availability

All of the code that I have written is available **on a private Github repository**. Here is a link to the repository: <https://github.com/hitesh05/Domain-Adaptive-NER>

I have kept it private for now since the project is still a work in progress, and I do not want anyone else to steal our code. I can always send a zip file to you in case you want to have a look at the code.

For more information, you can also have a look at the **working document** that I have maintained for this project:

<https://docs.google.com/document/d/1bNXKPrIoQNSyKoJcvZbrlhQTt-mhALnzNleWav3DBG8/edit?usp=sharing>

Qualitative Work

The qualitative work done during the semester involves reading up on various literature related to my research. I will provide a brief literature review of these texts and how they relate to the quantitative work I have been doing till now.

Book 1: The Gift by Marcel Mauss

Marcel Mauss, an influential anthropologist, delves into the intricate dynamics of gift exchange systems, challenging prevailing notions in anthropology during his time. His seminal work, "The Gift," critiques the binary classification of societies into simple and complex based on economic institutions, asserting that the concept of the "gift" holds crucial significance in understanding societal structures.

In the overarching theme of Mauss's work, the "gift" transcends a mere economic transaction; it is a "total social fact" imbued with spiritual mechanisms, intertwining the honour of both the giver and the receiver. By exploring the principle of reciprocity in primitive societies, Mauss navigates the labyrinth of obligatory and voluntary gifts, a phenomenon transcending temporal and cultural boundaries.

In Chapter 1, Mauss outlines the three main obligations inherent in a gift—giving, receiving, and returning—creating a cycle that mirrors the circulation of commodities. He emphasises the transdisciplinary nature of the gift, manifesting in economic, social, legal, and moral dimensions. Gifts, Mauss argues, act as agents binding individuals into moral relationships, with distrust arising when reciprocity is violated.

Chapter 2 extends Mauss's exploration to specific examples, such as the Samoan gift system and the Kula exchange. In Samoan culture, gifts hold spiritual essence, and their

non-reciprocation is deemed dangerous and immoral. The Kula exchange, akin to a social alliance, involves the exchange of gifts with symbolic and magical significance, revealing social hierarchies and class distinctions. Mauss introduces the concept of the Potlatch, a festive occasion where gifts are exchanged, signifying wealth, rank, and religious symbols. These rituals provide insights into the interplay of gifts, power, and social structures.

Mauss challenges the prevailing idea of a natural economy driven by self-interest, asserting that gifts form non-contractual contracts based on trust and faith. The act of giving and receiving establishes an implicit contract, emphasising relational continuity over economic transactionalism.

Hence, this work tells us how the gift is not separate from the market economy, as was believed before this research. He shows how the exchange of gifts ties intricately with the functioning of the economy in a society.

In conclusion, "The Gift" by Marcel Mauss emerges as a groundbreaking exploration of the intricate social, economic, and moral dimensions of gift exchange. It illuminates how gifts serve as instruments of communication, morality, and power, challenging conventional anthropological perspectives and emphasising the enduring relevance of these concepts across diverse societies and historical epochs.

Use in my research:

For my research, I can map the exchange of gifts from various periods and see if my findings correlate and overlap with the author, Marcel Mauss. As he has mentioned in the text, the circulation of gifts is similar to the exchange of commodities, and using computational tools, we can see if this hypothesis holds true. Also, the exchange of gifts has been said to take different forms amongst different classes of people. If we can map the gifts to the caste and the class of people involved in its exchange, we can once again try to verify whether this notion holds and whether we can figure out the existing social hierarchies in the societies.

Book 2: Trans-Himalayan Traders by James Fisher

James F. Fisher's anthropological study, "Trans-Himalayan Traders," offers a comprehensive exploration of the social, economic, and cultural intricacies within the isolated village of Tarangpur in northwest Nepal. Fisher's work unfolds across various chapters, each delving into specific aspects of the village's life and transactions, providing profound insights into the adaptive strategies of the community at the crossroads of diverse cultural influences.

Chapters 1 and 2: Cultural Hinge and Ethnographic Context

Fisher sets the stage by portraying Tarangpur as a cultural, linguistic, and economic hinge situated between the Buddhist Tibetan culture in the north and the Hindu Nepali region to the south and west. The village, nestled in the Dolpo District, emerges as a microcosm of cultural diversity within a confined geographical space. Fisher's approach is not merely

observational; it's participatory, navigating the challenges of research in a community suspicious of outsiders due to deeply ingrained beliefs in supernatural forces.

The linguistic tapestry of Tarangpur, encompassing Nepali, Tibetan, and the unique Kaike language, reflects the cultural amalgamation within the village. Fisher highlights the anthropological nature of the problems discussed, emphasising that the challenges are those of the anthropologist, not necessarily the people of Tarangpur, who are, in his view, well-adapted to their way of life and environment.

The geographical and cultural heterogeneity of Dolpo District is thoroughly examined, illustrating the cultural, linguistic, and ecological pluralism within the region. Fisher presents Tarangpur as a cultural bridge, linking the Tibetan Buddhist cultural tradition in the north with the caste Hindu-speaking Nepali culture in the south and southwest.

Chapter 4: The Salt Circuit

Fisher turns his attention to the economic heartbeat of Tarangpur – the salt trade. The chapter elucidates the intricate economic relationships between Tarangpurians and Bhotia traders who supply rock salt to the village. There is no natural source of salt in the region; hence, the people there depend upon the Bhotia traders to supply them with salt. They barter for salt in exchange for grain. The surplus salt is transported further south, where it is exchanged for rice, which is a luxury item in the region. Therefore, the grain-salt-rice exchange circuit emerges, and it is a vital economic and cultural transaction that weaves the villagers into an interconnected web with the external world. The people of Tarangpur thus have a special friendship with the Tarangpurian Bhotias in the north and with the Hundus in the south. An economic relationship forms the basis of this friendship, which is important for all the parties involved as their respective economies would find it hard to function without this particular grain-salt-rice circuit.

The people of Tarangpur coming into contact with different cultures creates a fascinating set of practices in the region. The Tarangpurians' engagement with the northern border area of Nepal and Tibetan Buddhism is marked by the adoption of Mahayana Buddhism, colloquially known as Lamasim. Their spoken and written language shifts to Tibetan, reflecting the profound influence of the Buddhist cultural tradition. This shift extends beyond mere linguistic adaptation, encompassing the embrace of Buddhist rituals, beliefs, and practices.

The description of the ankle-length dresses worn by women, full-length coats worn by men, and the unique rope and sole bags with wool loops extending to the knee exemplifies how the Tarangpurians not only adopt the religious aspects of Tibetan Buddhism but also integrate them into their daily lives through distinctive attire. On the flip side, the Tarangpurians' interaction with rice introduces them to the Hinduism lifestyle prevalent in the southern and southwestern regions of Nepal. Here, they navigate the complex landscape of caste groups, caste ranking, and the concept of parity inherent in Hindu society. The villagers showcase remarkable adaptability by practising impression management when immersed in Hindu areas, adopting Nepali language, and assuming entirely different names and identities. This adaptive strategy is not just linguistic but also embodies a different cultural persona.

The Tarangpurians occupy a unique position as cultural hinges between the two opposing cultural poles of the north and south. While commodities like salt and rice traverse the

cultural borders seamlessly, ideas encounter a stop in the middle pole. Hindu and Buddhist ideologies remain distinct, unaffected by each other. In their role as commercial middlemen and brokers of goods, the Tarangpurian traders not only facilitate the exchange of physical commodities but also act as ideological hinges. They adeptly navigate and practice both Hindu and Buddhist ideas in their respective areas, using the exchange of commodities as a means to ensure their survival. This dual engagement with different cultural, religious, linguistic, ecological, and social symbols positions them as key players in maintaining a delicate balance between two contrasting worlds.

Chapter 5: The Commodities Circuit

In the fifth chapter of "Trans-Himalayan Traders," James F. Fisher delves into the transformation of economic transactions within Tarangpur, emphasising the emergence of a second circuit that involves livestock, cloth, and manufactured goods. This evolution is marked by the strategic elimination of the middleman, showcasing the Tarangpurians' shrewd adaptability and resourcefulness.

As the traditional grain-salt-rice exchange circuit contracted, a new series of transactions gained prominence. The Tarangpurians demonstrated their ingenuity by bypassing middlemen, particularly merchants from Jumla in the west. Instead of relying on external traders, the villagers started directly engaging with the Terai region, thereby saving on transportation costs and avoiding expenses associated with supporting middlemen.

Fisher elucidates the dynamics of the livestock trade, which includes sheep, goats, and horses. This trade demands significant purchasing power, and the introduction of cash into the barter economy is explored. Traditionally, livestock, including those used for transporting salt and rice, would be bartered for grain. The practice of turning these animals into instant cash by selling them illustrates the Tarangpurians' financial acumen. The horse trade, in particular, is lucrative, yielding a substantial profit compared to sheep.

The woollen trade emerges as a significant economic activity, impacting a larger segment of the Tarangpurian population than the livestock trade. With minimal taxes on wool, this trade involves the transfer of resources between regions where wool is surplus and those where it is scarce. Men predominantly use knitting woollen products like sweaters, sashes, and blankets. The influence of Tibetan weaving traditions is evident in the style, variations, and workmanship that determine the cost of these products.

Tarangpurians navigate their way out of isolation by establishing economic and cultural ties that extend from Tibet to India. After trading livestock and woollen products in different middle hills, they convert the proceeds into goods by visiting various trade centres in India. This involves purchasing items like cigarettes, matches, soaps, cotton cloths, and Tibetan products, including tea, coats, gold, and jewellery. The surplus goods acquired are used at home, sold, or bartered in Bhot and Khasan.

Fisher draws a sharp contrast between the traditional grain-salt-rice exchange circuit and the evolving commodities circuit. The former addresses essential life requisites and continues in an attenuated form, remaining within the traditional, inter-regional symbiotic area. In contrast, the latter represents a shift to modern competitive markets, taking Tarangpurians

beyond their immediate cultural and geographical boundaries. This circuit is increasingly dominant, driven by a desire to satisfy various financial, ideological, and cultural needs.

This chapter portrays the Tarangpurians as astute cultural chameleons, adept at navigating different economic circuits and cultural contexts. Their ability to strategically eliminate the middleman, engage in diverse economic activities, and manage dual identities reflects economic resilience and a profound understanding of the complexities inherent in adapting to evolving trade dynamics. Fisher hints at the nuanced interplay between economic pursuits and cultural identity, paving the way for a deeper exploration in the subsequent chapter.

Conclusion:

Fisher's "Trans-Himalayan Traders" emerges as a tapestry rich in detail and nuance, weaving together the cultural, economic, and social threads of Tarangpurian life. Through meticulous ethnographic research, Fisher unveils the complexities of the villagers' interactions, transactions, and adaptive strategies. His work significantly contributes to understanding cultural intersections, economic resilience, and the intricate web of relationships in a Himalayan village. The book stands as a testament to the enduring importance of anthropological research in unravelling the multifaceted nature of human societies.

Use in my research:

This work could once again be fascinating to corroborate using computational tools. If we can map the salt and the commodities circuit, we can analyse the exchange of not just goods and commodities but also the amalgamation of cultures in various regions of the Western Himalayas. The use of computational tools would also mean that we can scale our work not just spatially but also temporally, allowing us to analyse these trade routes and exchange of commodities and ideas on a large scale. This could give us an idea of not just the differences in how the commodities travelled in various places but also how these exchanges varied over time.

Book 3: Caravans by Scott Levi

"Caravans" by Scott Levi unfolds a captivating narrative that delves into the remarkable journeys Multani and Shikarpuri merchants embarked upon from the sixteenth to the nineteenth centuries. In contrast to the perception of caravan traders as mere peddlers, Levi sheds light on the sophisticated techniques employed by these intrepid individuals, illustrating how they transformed modest merchandise into extensive trade and moneylending ventures.

The book challenges the prevailing notion that the surge of European trade in the Indian Ocean displaced overland "Silk Road" commerce. Instead, Levi argues that Multani traders played a pivotal role in bridging the early modern Indian and Central Asian economies. He traces the intricate trade network originating in Multan, where Indian colonies were established in foreign lands. These colonies served as pivotal points for caravans to

replenish stocks of commodities like cotton textiles, indigo, sugar, rice, spices, weapons, and precious stones. In return, copious amounts of gold and silver flowed into India.

A key focus of the book is the intertwined nature of the slave trade and the horse trade. India, known for its cultivation of cotton and textile production since antiquity, not only exported goods but also slaves. Horses were a significant commodity, with Indian rulers importing them from Central Asia. Levi vividly portrays the intricate network of caravan routes, emphasising that the exchange between traders was not solely economic but also cultural. The spread of Buddhism along these trade routes underscores the profound impact of these journeys. The narrative touches upon the diaspora of Multani and Shikarpuri traders, exploring how disruptions during Nadir Shah's reign led to a decline in the Indian merchant presence. Despite challenges, these traders adapted to changing circumstances and continued their activities, establishing themselves in various regions.

Now, I will discuss about the **chapter on “Indian Textiles”** in the book. This chapter delves into the significant role of Indian textiles in the intricate web of trade along the caravan routes. While references to luxury goods like silk and furs are abundant due to their association with the affluent, most merchandise traversing the caravan routes comprised bulk goods. Cotton textiles emerged as a central focus, constituting India's most crucial export during the early modern era. The caravan routes facilitated the movement of various commodities, with textiles taking the lead, followed by indigo, other dyes, sugar, rice, jewellery, and spices. The trade in textiles was not confined to a specific region but spanned across Bengal, the Coromandel Coast, Gujarat, and the north-western regions like Punjab and Multan.

European accounts from the 17th century shed light on the substantial volume of Indian merchandise transported via camels along overland trade routes, particularly from Qandahar to Isfahan. The references encompass various Indian textiles in Central Asian markets, including Kashmiri shawls, handkerchiefs, turbans, towels, and robes. Lahore emerges as a pivotal commercial centre, receiving the bulk of cotton production in the region. This city plays a crucial role in trade with Uzbek Central Asia. As highlighted in the text, goods purchased in Lahore find their way to the Uzbeks or Tartars, often transported via Kabul to various destinations.

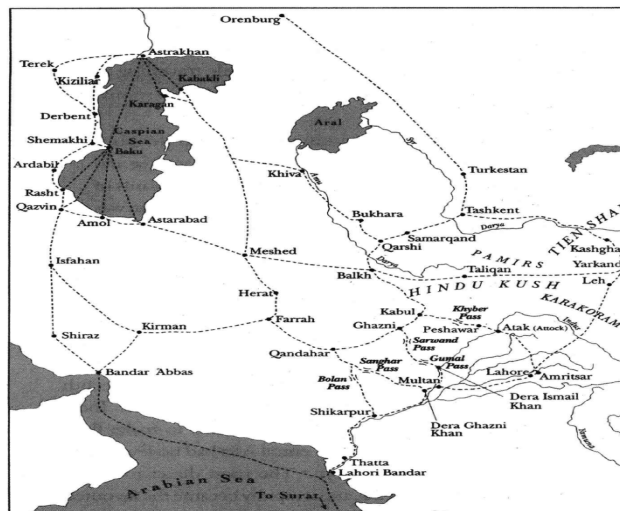
The chapter emphasises that while some items reached Central Asian markets for local consumption, many were part of a vast transit trade. Indian products moved beyond Central Asia, reaching markets in Siberia, China, Muscovy, and beyond. The trade routes served as conduits for the exchange of goods, with Indian textiles playing a significant role. The text suggests that textiles, including those transported by Bukharan traders, likely originated in India. This dynamic exchange of goods highlights the multifaceted nature of the caravan trade. Indian textiles not only met local demands but also became integral to the broader transcontinental trade networks, connecting diverse regions and cultures. The chapter underscores the pivotal position of textiles in shaping the economic and cultural ties along the caravan routes.

The book also addresses the repercussions of Russian expansion on Indian presence in Central Asia during the latter half of the nineteenth century. Harsh restrictions and policies contributed to declining Indian businesses in Russian territories.

In conclusion, "Caravans" paints a vivid picture of the resilience, adaptability, and economic prowess of Multani and Shikarpuri merchants. It highlights their crucial role in shaping trade dynamics between India and Central Asia, leaving a lasting legacy. Examples such as the Hinduja family's successful global business originating in Shikarpur serve as testaments to the enduring impact of these enterprising traders.

Use in my research:

Here is a map of the principal trade routes provided in the text:



Map of the Principal Trade Routes
(Not to scale)

I can use my algorithm to draw similar trade routes from the references provided in their research and see how they overlap with the given maps. This will serve not only as a way to test the hypothesis of the book but also to see how well our algorithm performs for a given set of data since, in a way, we also have the gold translation available.

Book 4: The Art Of Not Being Governed by James Scott

James C. Scott's "The Art of Not Being Governed" redefines our understanding of the trajectory of modern civilisation, challenging the prevailing narrative of inevitable statehood. Focused on the rugged region of Zomia, Scott introduces the concept that the diverse peoples inhabiting this area are "stateless by choice."

Zomia emerges as a refuge actively sought by those who wish to evade state control. It is a vast area of Southeast Asia characterised by rugged terrain and inhabited by various highland peoples. Scott dismisses the notion that these communities are remnants of a Neolithic era, arguing instead that they intentionally fled lowland states over the last 2,000 years. Drawing on evidence from ecology, archaeology, linguistics, history, and ethnography, the book challenges the stereotype of Zomians as civilisations left behind, emphasising that they are individuals who choose to escape.

This intentional avoidance of assimilation into lowland states by the "hill peoples" of Zomia forms a central theme in Scott's analysis. Embracing an anarchistic lens, the book highlights the resilience and ingenuity of these societies in maintaining autonomy. Scott delves into the civilising missions of states and the active resistance met by upland peoples keen on preserving their unique ways of life.

Zomia resists state control due to its challenging geography—mountains, plateaus, and dense forests. These geographical features have historically thwarted attempts by states to assert authority, rendering Zomia a "shatter zone" — a refuge from state dominance.

The strategic choice of "hill peoples" to inhabit mountains and resist state control becomes apparent as Scott explores the contrast between the swidden agriculture practised by these communities, allowing for mobility, and the sedentary lifestyle of valley peoples practising wet rice cultivation.

Central to Scott's argument is the concept of the "art of not being governed," which encompasses various strategies employed by upland peoples to resist state control. These strategies include:

1. Physical mobility: Upland societies are often characterised by their mobility in terms of geographic location and social organisation. They can quickly move their villages to avoid state encroachment.
2. Cultivation Practices: Swidden agriculture, or shifting cultivation, allows for land to be cultivated and then left fallow, preventing the establishment of permanent settlements that would be easier for states to control.
3. Language: Upland groups often maintain distinct languages that isolate them from lowland societies, making communication and control difficult.
4. Religious Practices: Religion, particularly animism, is an integral part of upland cultures and can serve as a source of identity and resistance.

Scott explores the physical and social barriers that prevent state control in upland regions. The challenging geography and the development of social mechanisms within upland communities make it difficult for states to impose authority.

Within this context, perceived "outlaws" in upland societies play a crucial role in resisting state authority, contributing to preserving autonomy. Upland peoples may also feign loyalty to the state while privately preserving their true traditions, navigating state rules, and safeguarding their autonomy.

Scott explicitly discusses the book's connection to anarchist thought, emphasising the anarchistic characteristics of upland societies that prioritise autonomy and resist centralised authority. The conclusion summarises major themes and arguments, underscoring the ongoing relevance of upland societies' strategies for resisting state control.

In essence, "The Art of Not Being Governed" challenges the conventional view that states are all-encompassing, presenting a paradigm shift by highlighting the agency and ingenuity of upland peoples who strategically avoid state control. Scott's work adds a nuanced layer to understanding human history and the complex relationship between statehood and autonomy.

Use in my research:

We can map the movement of these people and see if we are able to differentiate the region of Zomia from the rest. We can also do a temporal analysis of their movements and see if there are any patterns that emerge that help us understand these societies better. Mapping their cultivation practices and the places where they practise their agriculture could be one way to characterise their movements.

Book 5: A Geographical Introduction to History by Lucien Febvre

For now, I have only read chapters 1 and 2 of part four of the book, and I will be writing about these. I shall continue to read this text throughout my winter vacations.

Febvre comments on how mountain ranges and rivers are often considered to form natural frontiers and boundaries for states, but most of these are actually intellectual creations of man. He critiques our tendency to look at the past and the geography of the past with the eyes of the present. Thus, mountains and rivers form borders that didn't actually exist at the time. More so, these frontiers are often considered simply boundaries and are studied as such, not as zones where people settle and live. This, too, is a tendency that has to be overcome.

Communication and routes are also determined through geography but also influenced by man's adaption. Rivers can be obstacles or markers for a route depending on a number of factors, starting from how men use them and what weather it is like. Some routes fall out of favour at a time and are back in favour later. The reason great trade routes are often used again and again however, is out of necessity; it is because of the commercial advantages that new routes are found and old routes continue to be in use. Thus the value of the route does not necessarily lie in the track followed, but in how much man needs it. The nature of traffic which feeds a route is paramount to other considerations. If there is a need for communication, man will find a way to overcome the geographical limitations.

Use in my research:

The author also characterises two types of soil in the text: political and geographical. While I have not read this topic in detail, the author discusses how these two soils are generally separate and may not always overlap. Once again, I can use my algorithm to try to map the regions marked by these soils and corroborate whether what the author has said holds for our dataset or not.

Book 6: Becoming India by Aniket Alam

Chapter 1 of "Becoming India" by Aniket Alam focuses on the geography of the Western Himalayas and its significance as a strategic line of defence and a physical boundary between India and its neighbouring regions. The Western Himalayas are dissected into three distinct zones: Western, Central, and Eastern. Among these, the Sutlej River stands out as a crucial watercourse that traverses the Himalayan terrain, playing a pivotal role in trade within the mountainous regions.

The chapter elaborates on the division of the Western Himalayas into parallel ranges, comprising the Outer Himalayas (Shiwalik), Lesser Himalayas (Central Zone), and the Great Himalayas and Zaskar (Northern Zone). The isolation of river valleys in this region results in distinct ecological and social structures for each valley. Despite similarities, these valleys have had limited contact, leading to variations in relief features, habitation patterns, and historical links.

The Himalayan climate exhibits significant diversity influenced by the folds of the mountains, ridges, and valleys. The region's soils are characterised as thin and unstable overall, but larger valleys witness soil accumulation on slopes. Vegetation in the Himalayas primarily consists of two types of forests: deciduous and coniferous. The chapter categorises plants based on their economic significance, emphasising the varying levels of importance.

Examining demographic aspects, the Western Himalayas exhibit a low sex ratio. Population density decreases as one moves deeper into the Himalayan range. The dominant presence of agricultural castes characterises the region, with valleys hosting the greatest concentration of human population. Different settlement types, varying in isolation and fragmentation, coexist in the area. Agriculture and pastoralism emerged as the primary occupations, with river valleys serving as crucial communication routes.

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